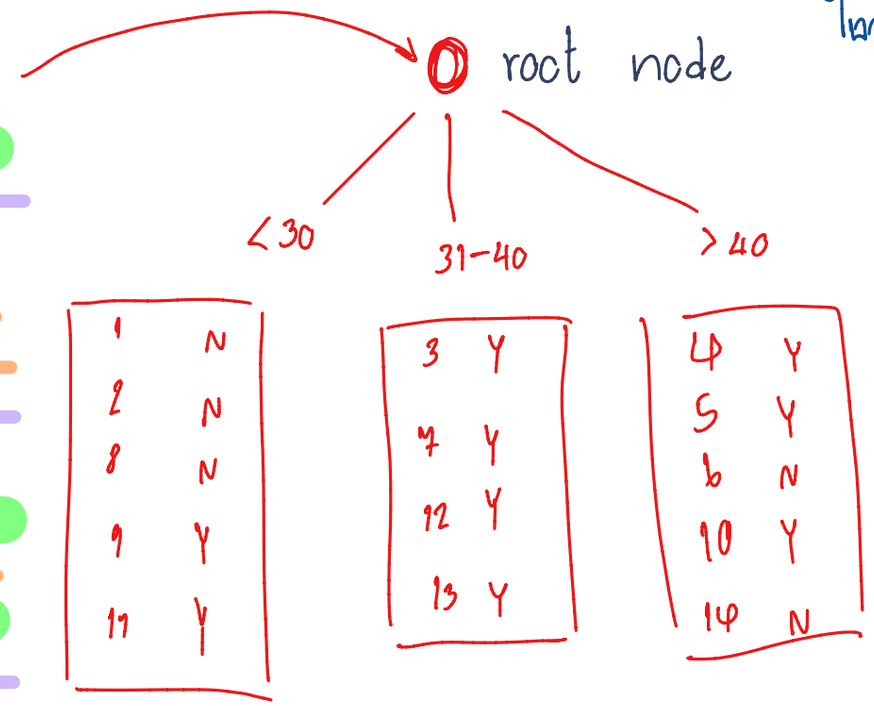
ไพรินทร์ ชื่นแสน 663020587-2

	age	income	student	credit_rating	buys_computer
1	<=30	high	no	fair	no
2	<=30	high	no	excellent	no
3	31…40	high	no	fair	yes
4	>40	medium	no	fair	yes
5	>40	low	yes	fair	yes
6	>40	low	yes	excellent	no
7	31…40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
9	<=30	low	yes	fair	yes
10	>40	medium	yes	fair	yes
11	<=30	medium	yes	excellent	yes
12	31…40	medium	no	excellent	yes
13	31…40	high	yes	fair	yes
14	>40	medium	no	excellent	no



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6	>40	low	yes	excellent	no
7	31…40	low	yes	excellent	yes
8	<=30	medium	no	fair	no
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12	31…40	medium	no	excellent	yes
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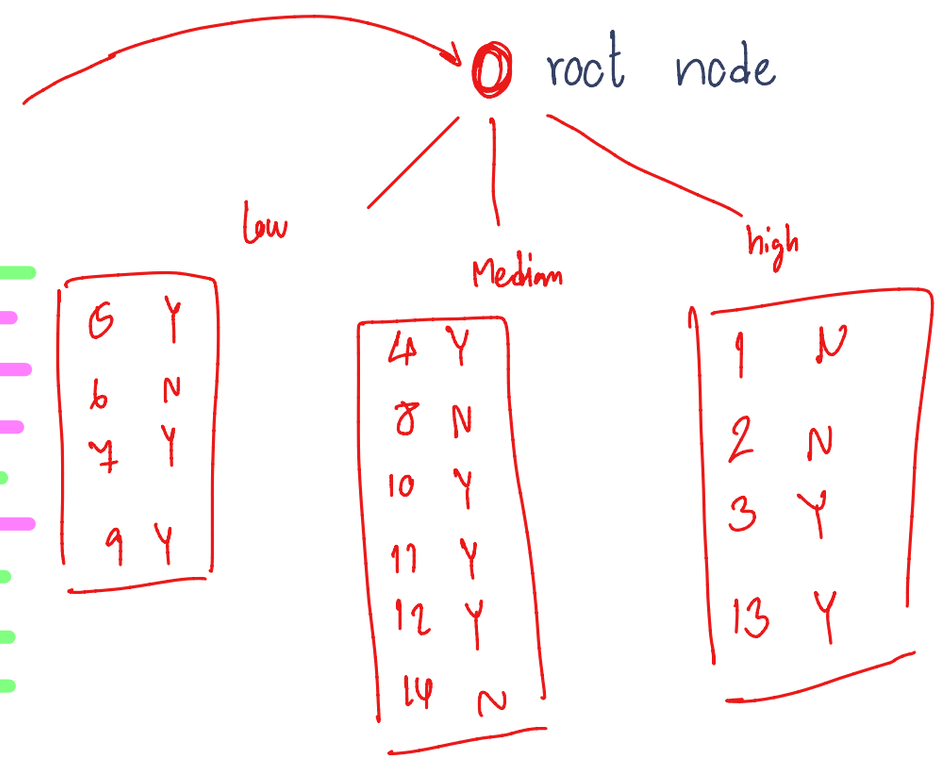
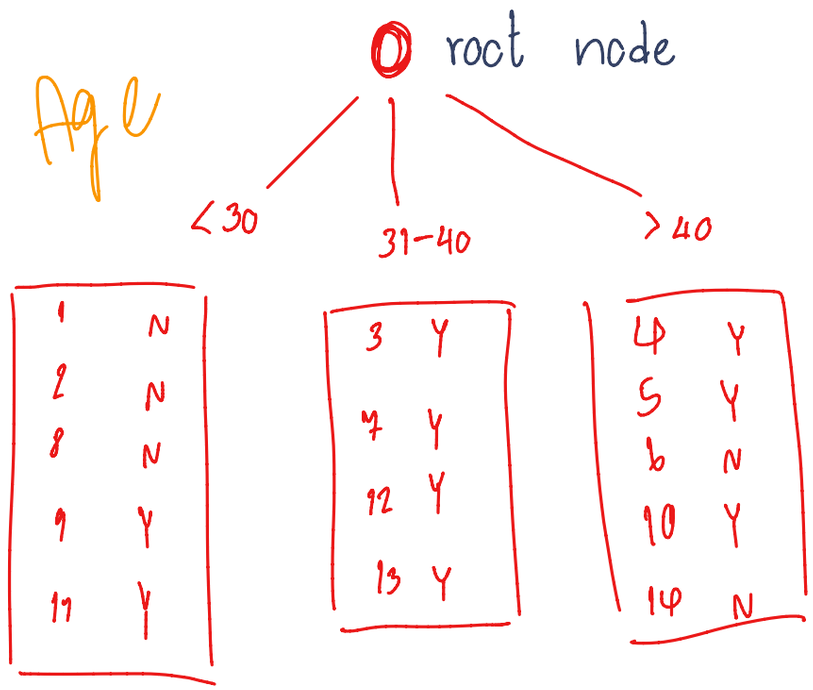
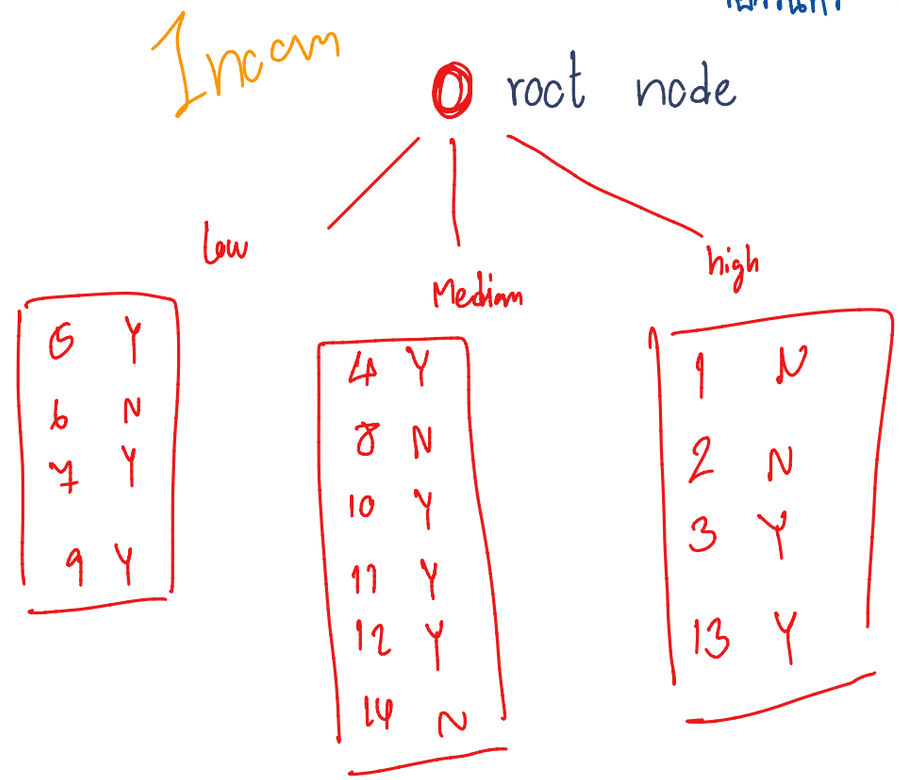
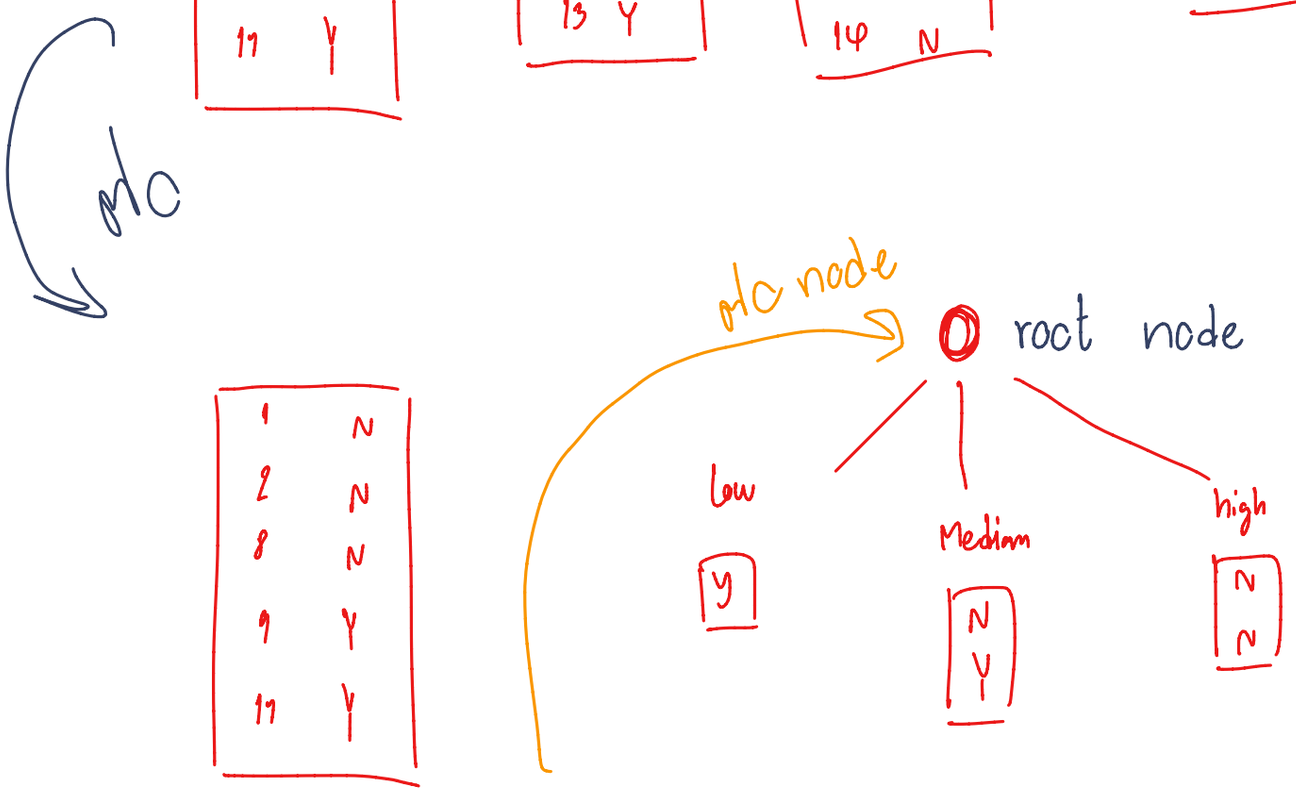


ไพรินทร์ ชื่นแสน 66302058 7-2



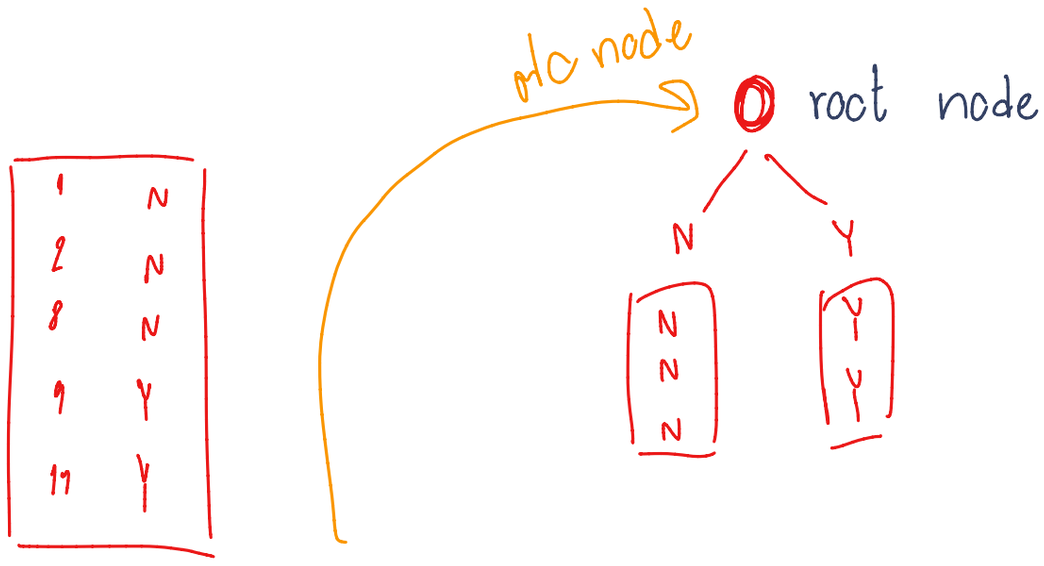


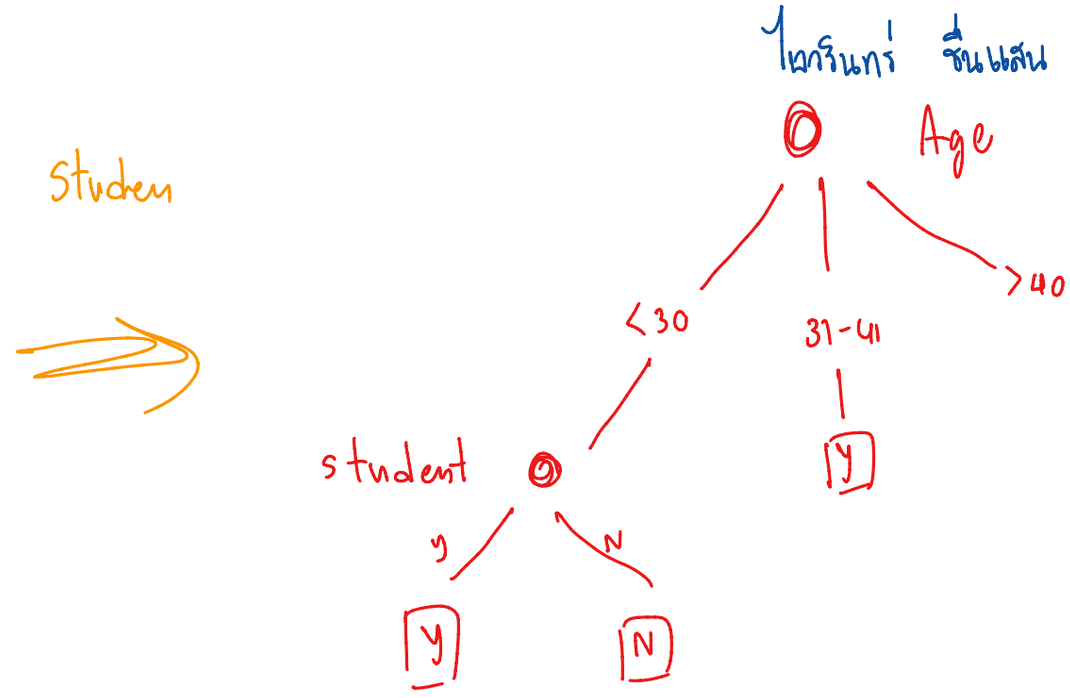


age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
<=30	medium	yes	excellent	yes

age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
31…40	high	no	fair	yes
>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
>40	medium	no	excellent	no

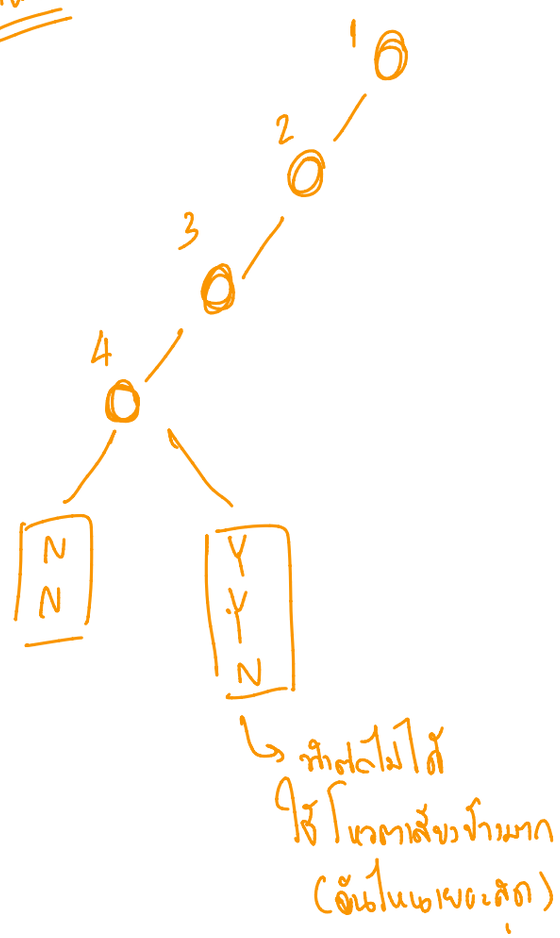
ไพรินทร์ ชื่นแสน 663020587-2



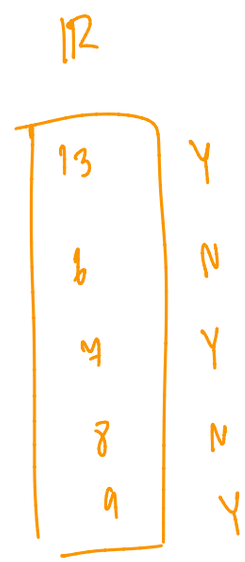


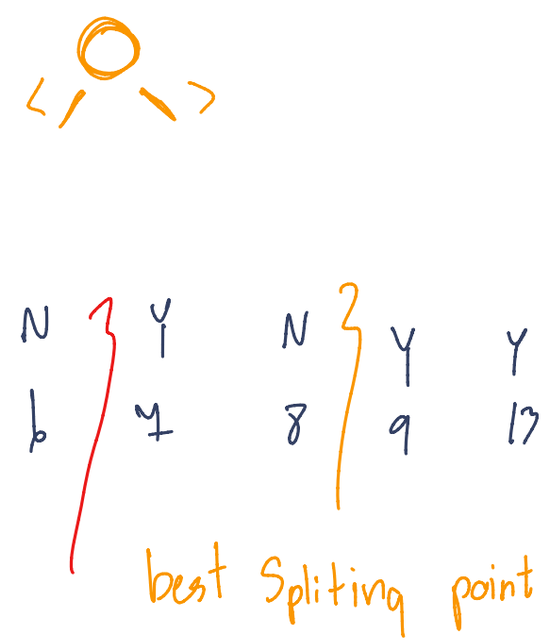
age	income	student	credit_rating	buys_computer
<=30	high	no	fair	no
<=30	high	no	excellent	no
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
<=30	medium	yes	excellent	yes

กรณี 1



กรณี 2





ไพรินทร์ ชื่นแสน 66302058 7-2

age	income	student	credit_rating	buys_computer
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<=30	high	no	excellent	no
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>40	medium	no	fair	yes
>40	low	yes	fair	yes
>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
<=30	medium	no	fair	no
<=30	low	yes	fair	yes
>40	medium	yes	fair	yes
<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
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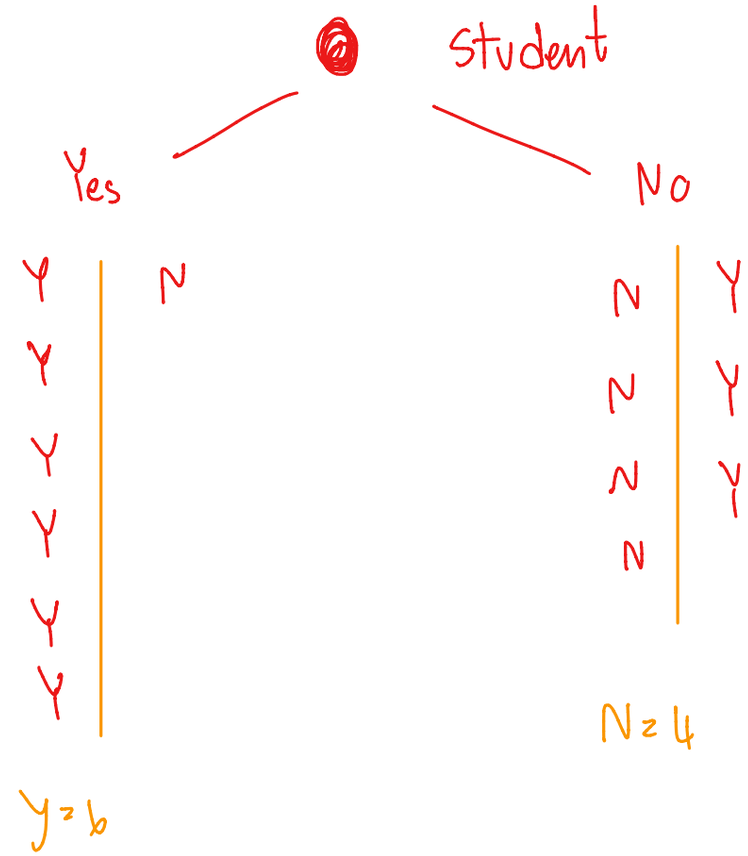

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<=30	low	yes	fair	yes
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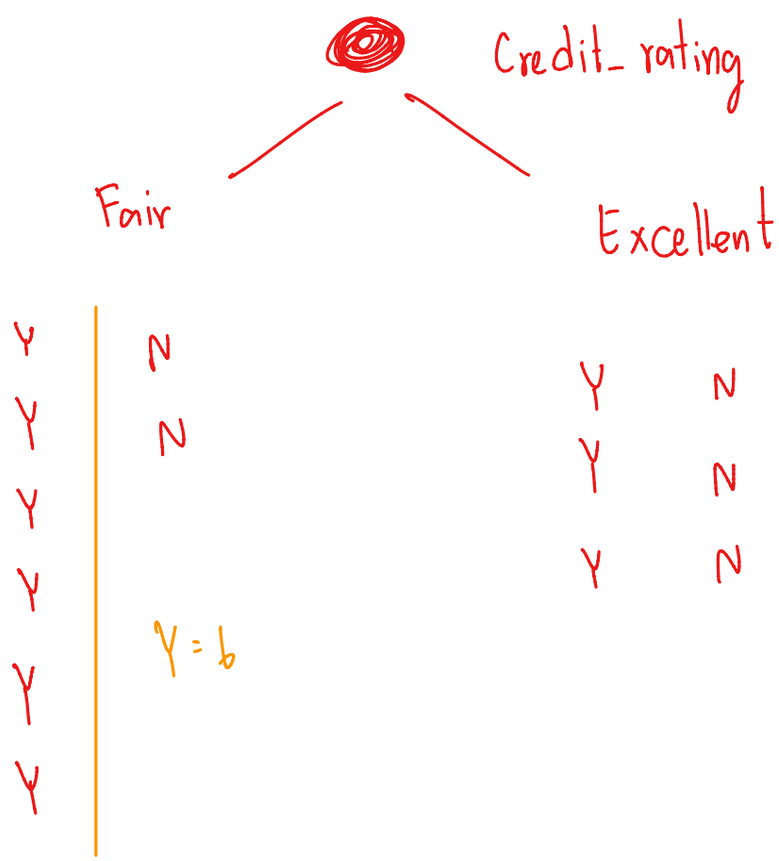

HW

ไพรินทร์ ชื่นแสน 663020587-2

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>40	low	yes	excellent	no
31…40	low	yes	excellent	yes
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<=30	low	yes	fair	yes
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<=30	medium	yes	excellent	yes
31…40	medium	no	excellent	yes
31…40	high	yes	fair	yes
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1.) age

จะได้ $\text{Gain}(A) = \underbrace{\text{Info}(D)}_{\text{ก่อนจัก}} - \underbrace{\text{Info}_{age}(D)}_{\text{หลังจัก}}$

หา $\text{Info}(D)$

$$= I(9,5) = \frac{-9}{14}\log_2\left(\frac{9}{14}\right) - \left(\frac{5}{14}\right)\log_2\left(\frac{5}{14}\right) = 0.940$$

หา $\text{Info}_{age}(D)$

$$= \underbrace{\frac{5}{14} I(2,3)}_{\leq 30} + \underbrace{\frac{4}{14} I(4,0)}_{30-40} + \underbrace{\frac{5}{14} I(3,2)}_{>40}$$

$$= \frac{5}{14}\left[\frac{-2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right)\right] + \cancel{\frac{4}{14} I(4,0)}^{\,0} + \frac{5}{14}\left[-\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right)\right] = 0.694$$

$(3,1) = 4$

$(4,2) = 6$

$(2,2) = 4$

จะได้

age	p_i	n_i	$I(p_i, n_i)$
≤ 30	2	3	0.971
30-40	4	0	0
>40	3	2	0.971

$\therefore \text{Gain}_{age} = 0.940 - 0.694 = 0.246$ #

ไพศิษฐ์ ชื่นชสน 663020587-2

2.) Income

$$Gain_{income} = Info(D) - Info_{income}(D)$$

$$Info_{income}(D)$$

$$= \frac{4}{14} I(3,1) + \frac{6}{14} I(4,2) + \frac{4}{14} I(2,2)$$

$$= \frac{4}{14}\left[-\frac{3}{4}\log_2\left(\frac{3}{4}\right) - \frac{1}{4}\log_2\left(\frac{1}{4}\right)\right] + \frac{6}{14}\left[-\frac{4}{6}\log_2\left(\frac{4}{6}\right) - \frac{2}{6}\log_2\left(\frac{2}{6}\right)\right] + \frac{4}{14}\left[-\frac{2}{4}\log_2\left(\frac{2}{4}\right) - \frac{2}{4}\log_2\left(\frac{2}{4}\right)\right]$$

$$= 0.232 + 0.394 + 0.285$$

$$= 0.911$$

$$\therefore Gain_{income} = 0.940 - 0.911 = 0.029 \#$$

3. Student

$$Gain_{student} = Info(D) + Info_{student}(D)$$

$$Info_{student}(D)$$

$$= \frac{7}{14}\left[-\frac{6}{7}\log_2\left(\frac{6}{7}\right) - \frac{1}{7}\log_2\left(\frac{1}{7}\right)\right] + \frac{7}{14}\left[-\frac{3}{7}\log_2\left(\frac{3}{7}\right) - \frac{4}{7}\log_2\left(\frac{4}{7}\right)\right]$$

$$= 0.296 + 0.493$$

$$= 0.789$$

$$\therefore Gain_{student} = 0.940 - 0.789 = 0.151 \#$$

ไพศรินทร์ ชื่นชื่น 663020587-2

4.) Credit_rating

$$Gain_{credit} = Info(D) - Info_{credit}(D)$$

หา $Info_{credit}(D)$

$$= \frac{8}{14} I(6,2) + \frac{6}{14} I(3,3)$$

$$= \frac{8}{14}\left[-\frac{6}{8}\log_2\left(\frac{6}{8}\right) - \frac{2}{8}\log_2\left(\frac{2}{8}\right)\right] + \frac{6}{14}\left[-\frac{3}{6}\log_2\left(\frac{3}{6}\right) - \frac{3}{6}\log_2\left(\frac{3}{6}\right)\right]$$

$$= 0.464 + 0.429$$

$$= 0.893$$

$$\therefore Gain_{credit} = 0.940 - 0.893 = 0.047$$